

## SAFETY DATA SHEET

Revision Date 30-March-2024

Revision Number 3

### 1. Identification

**Product Name** Molybdenum(III) chloride

**Cat No. :** 14034

**CAS-No** 13478-18-7  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Causes severe skin burns and eye damage

May cause respiratory irritation

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                | CAS-No     | Weight % |
|------------------------------------------|------------|----------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | <=100    |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                          |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Ingestion</b>                       | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                               |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                              |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                        |

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Not combustible. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Hydrogen chloride. Molybdenum oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

**Environmental Precautions** Should not be released into the environment. Do not allow material to contaminate ground water system. See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong bases. Oxidizing agent.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                                | Alberta                                                                             | British Columbia                                                                    | Ontario TWAEV                                                                       | Quebec                                                                              | ACGIH TLV                                                                           | OSHA PEL                                                                  | NIOSH                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup><br>TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) TWA: 10 mg/m <sup>3</sup> | IDLH: 1000 mg/m <sup>3</sup> IDLH: 5000 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location.

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Solid                    |
| <b>Appearance</b>                             | Dark red                 |
| <b>Odor</b>                                   | Acidic                   |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | No information available |
| <b>Melting Point/Range</b>                    | No data available        |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| <b>Upper</b>                                  | No data available        |
| <b>Lower</b>                                  | No data available        |
| <b>Vapor Pressure</b>                         | No information available |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Specific Gravity</b>                       | No information available |
| <b>Solubility</b>                             | No information available |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |

|                           |                          |
|---------------------------|--------------------------|
| Decomposition Temperature | No information available |
| Viscosity                 | Not applicable           |
| Molecular Formula         | MoCl <sub>3</sub>        |
| Molecular Weight          | 202.32                   |

## 10. Stability and reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactive Hazard                  | None known, based on information available |
| Stability                        | Air sensitive. Moisture sensitive.         |
| Conditions to Avoid              | Incompatible products.                     |
| Incompatible Materials           | Strong bases, Oxidizing agent              |
| Hazardous Decomposition Products | Hydrogen chloride, Molybdenum oxides       |
| Hazardous Polymerization         | Hazardous polymerization does not occur.   |
| Hazardous Reactions              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                | CAS-No     | IARC       | NTP        | ACGIH | OSHA       | Mexico     |
|------------------------------------------|------------|------------|------------|-------|------------|------------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | Not listed | Not listed | A3    | Not listed | Not listed |

ACGIH: (American Conference of Governmental Industrial Hygienists)

A1 - Known Human Carcinogen  
A2 - Suspected Human Carcinogen  
A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

|                          |                           |
|--------------------------|---------------------------|
| Mutagenic Effects        | No information available  |
| Reproductive Effects     | No information available. |
| Developmental Effects    | No information available. |
| Teratogenicity           | No information available. |
| STOT - single exposure   | Respiratory system        |
| STOT - repeated exposure | None known                |
| Aspiration hazard        | No information available  |

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                        |                                                                |
|----------------------------------------|----------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | No information available                                       |
| <b>Other Adverse Effects</b>           | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and Degradability</b> | May persist               |
| <b>Bioaccumulation/ Accumulation</b> | No information available. |
| <b>Mobility</b>                      | No information available. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Technical Name</b>       | (Molybdenum(III) chloride)                 |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |

### TDG

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |

### IATA

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |

### IMDG/IMO

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, inorganic, n.o.s. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |

## 15. Regulatory information

### International Inventories

| Component                                | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | -   | X    | X    | ACTIVE                                        | 236-775-9 | -      | -   |

| Component                                | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | -     | KE-25461 | -    | -    | X    | -    | -     | -     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances  
ENCS - Japanese Existing and New Chemical Substances  
AICS - Australian Inventory of Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

Authorisation/Restrictions according to EU REACH Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Molybdenum chloride (MoCl <sub>3</sub> ) | 13478-18-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 30-March-2024  
**Print Date** 30-March-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**